

B.Sc. (Hons.) Semester IV Examination 2022-23
STATISTICS

Paper No. STB-401: Sample Surveys and Design of Experiments

Time: Three hours

Max. Marks: 70

NOTE: Attempt any five questions. All questions carry equal marks.

1. (a) The households in a town are to be sampled in order to estimate the average amount of assets per household. The households are stratified into a high-rent and low-rent stratum. A house in the high-rent stratum is thought to have about nine times as much assets as one in the low-rent stratum, and S_i is expected to be proportional to the square root of the stratum mean. There are 4,000 households in the high-rent stratum and 20,000 in the low-rent stratum. Distribute a sample of 1,000 households between the two strata.
(b) In a stratification with two strata, given $W_1 = 0.8$, $W_2 = 0.2$, $S_1 = 2$, and $S_2 = 4$, compute the sample sizes n_1 and n_2 ignoring fpc if the standard error of the estimated population mean $\bar{y}_{st}$ is to be 0.1 and the total sample size $n = n_1 + n_2$ is to be minimized.
 2. Discuss in detail the different types of errors involved in sampling, highlighting the distinctions between sampling and non-sampling errors.
 3. (a) Consider a population of 6 units with values 1, 2, 3, 4, 5, 6. Write down all possible samples of size 2 (without replacement) from this population and verify that the sample mean is an unbiased estimate of the population mean. Also calculate its sampling variance and verify that:
 - i. it agrees with the formula for the variance of the sample mean, and
 - ii. this variance is less than the variance obtained from sampling with replacement.(b) Discuss the process of determination of sample size for obtaining an estimate with a specified coefficient of dispersion.
 4. Discuss the ratio, regression, and product methods of estimation. Mention their conditions of preference and importance in sampling theory.
 5. Describe the Analysis of Variance (ANOVA) technique for a fixed-effect model for one-way classified data. Also discuss the use and method of obtaining the critical difference (CD) in this context.
 6. (a) Mention the advantages and disadvantages of Completely Randomized Design (CRD), Randomized Block Design (RBD), and Latin Square Design (LSD).
(b) Explain the mathematical procedure of estimating one missing observation in a Randomized Block Design (RBD).
 7. What are the basic principles of design of experiments? Explain each with suitable examples. Also discuss the formal layout and statistical model of a Latin Square Design (LSD).
 8. Write descriptive comparisons explaining the advantages of:
 - (a) simple random sampling over systematic sampling
 - (b) Latin Square Design (LSD) over Randomized Block Design (RBD)
 - (c) objective sampling over subjective sampling
 - (d) sample survey over complete enumeration
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- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app/>